

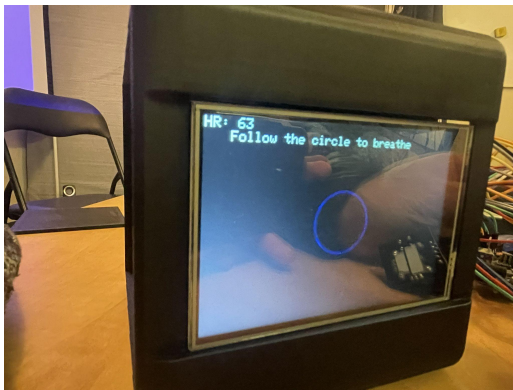
Jeffrey Xu Design Portfolio

Portable Sensory Cube

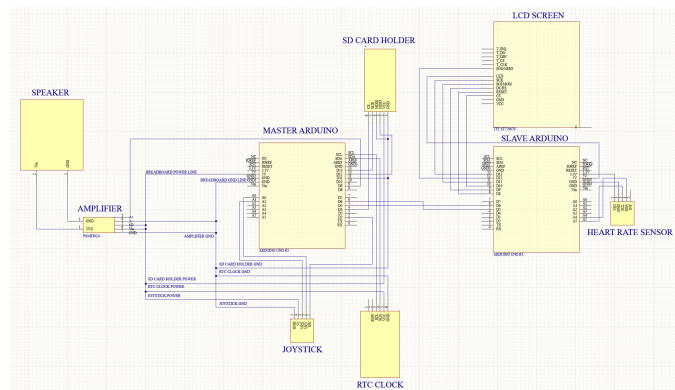
ARDUINO MEDICAL ASSISTIVE DEVICE

An Arduino-based assistive device aimed at mitigating hospital-induced delirium. It drives a TFT LCD over SPI to guide breathing exercises, and reads a MAX30102 pulse oximeter over I2C — configuring the sensor's registers to sample real-time photoplethysmography (PPG) data for heart-rate processing. Housed in a 3D-printed chassis with a joystick, speaker, and RTC clock.

[Arduino] [C] [SPI] [I2C] [MAX30102]



finished device



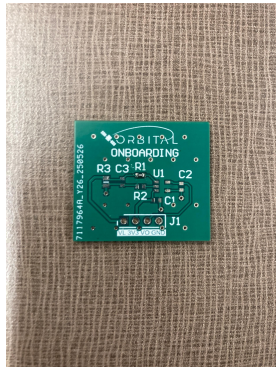
wiring schematic

IC Current-Sense Amplifier

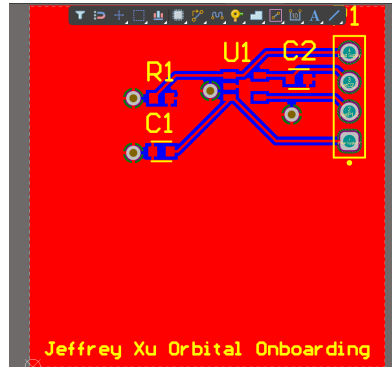
UW ORBITAL / SATELLITE EPS HARDWARE

A current-sense circuit for UW Orbital's satellite power system, built around an INA180 amplifier and a $0.22\ \Omega$ shunt resistor sized to limit power loss. I followed the INA180 datasheet layout guidance — checking capacitor voltage ratings and shunt power rating — to improve measurement stability and cut noise in the sense path, then laid out and soldered the PCB in Altium Designer.

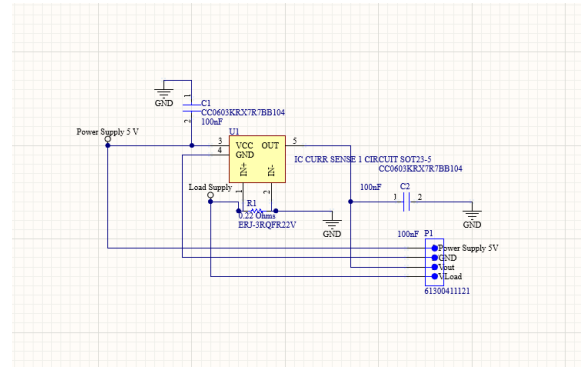
[Altium Designer] [INA180] [Analog Design] [PCB Layout] [Soldering]



soldered PCB



Altium layout



schematic